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Experimental Investigation into the Influence of Infill Density, Print Pattern, and Built-Up Direction on the Flexural Strength of FFF-Manufactured PLA Components

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Abstract

This study evaluates the flexural strength of poly lactic acid parts (PLAs) fabricated with fused filament fabrication (FFF) by systematically analyzing the combined effects of infill density, infill pattern, and built-up orientation. Therefore, samples with 10, 30, 50, 70, and 90% infill densities were printed with cubic and triangular patterns in all three possible built-up directions (Cartesian X, Y, Z) and subjected to a standardized three-point bending test according to ISO 178, while printing time was concurrently assessed to quantify trade-offs between mechanical performance and manufacturing efficiency. The results show that a cubic infill with layers transverse to the bending load (Y-direction) offers the highest flexural strength of about 31 MPa for 90% infill density at comparably low printing times. In addition to significantly longer printing times, samples printed in the X-direction achieved the highest flexural strengths across all configurations tested for both infill patterns examined, up to densities of approximately 60%.

Keywords: additive manufacturing; fused filament fabrication (FFF); 3D-print; flexural strength; bending test; anisotropy; infill density; print pattern

1. Introduction

Additive Manufacturing (AM), commonly known as 3D printing, has revolutionized product development and manufacturing by enabling the fabrication of complex geometries with minimal material waste and shortened production cycles. Among the available AM techniques, fused filament fabrication (FFF) has gained particular importance due to its cost-effectiveness, simplicity, and accessibility for polymer-based components [1,2]. In FFF, a thermoplastic filament is extruded layer by layer to form three-dimensional structures (Figure 1), allowing for high customization of internal parameters, such as infill density, pattern, and built-up direction. These parameters directly influence the mechanical response and anisotropic behavior of the printed parts [3,4]. The mechanical performance of FFF-manufactured components is inherently anisotropic, primarily due to the directional nature of the layer deposition process. The bonding quality between adjacent layers and the alignment of filaments govern critical properties such as tensile and flexural strength, stiffness, and failure modes [3,5,6]. Early studies have demonstrated that horizontally printed samples typically outperform vertically built ones in terms of load-bearing capacity due to superior interlayer adhesion and reduced stress concentration at interfacial boundaries [4,7–9]. Furthermore, variations in printing parameters—such as

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layer height, raster angle, and infill geometry—significantly affect the structural integrity and energy absorption behavior of the parts [10,11].

For samples manufactured using MEX sintering (material extrusion sintering), it has been shown that the built-up direction and raster angle of the green parts have influence on the flexural strength of the finished metal sample, as ref. [12] demonstrate.

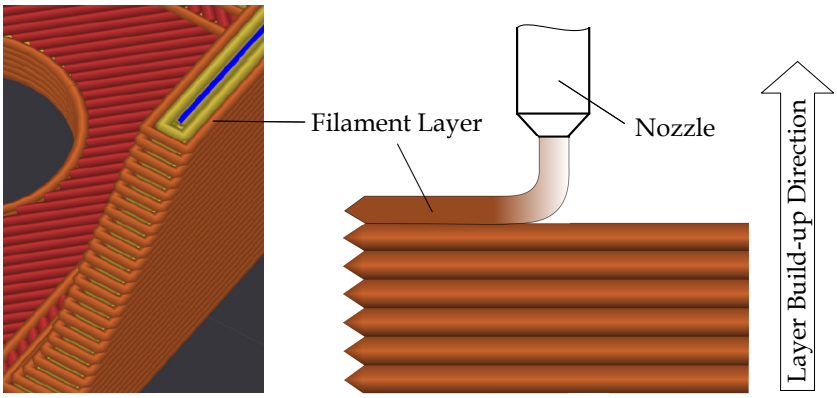


Figure 1. Scheme of the fused filament fabrication process [9].

Infill density and pattern play particularly crucial roles in tailoring the stiffness, weight, and failure mechanisms of printed thermoplastic components. Studies show that advanced infill patterns—such as gyroid or other continuous-surface structures—can significantly enhance strength-to-weight efficiency and delay crack initiation compared to conventional rectilinear infills [13]. Variable-density strategies further improve load distribution by locally tailoring stiffness and energy absorption, leading to a superior mechanical response under complex loading [14]. Moreover, frequency-dependent fatigue investigations confirm that the interplay between printing parameters and infill architecture critically affects cyclic durability, with optimized raster organization and infill morphology markedly extending fatigue life [15].

Triangular, hexagonal, and cubic infill structures have been widely studied for their ability to modify stress distribution and enhance the specific strength of printed materials [16–20]. In [20], the focus of the investigations was not only on the printing direction but also on the influence of the raster angle on the flexural properties of ABS samples.

Other studies have also extensively investigated the influence of infill density and built-up direction on flexural strength using three-point bending tests, but their methods differ from those used in this study in that they only examined one of three possible built-up directions [21,22] or without variation in infill density [18–20,22]. The influence of parameter variation on printing time in combination with flexural strength is also not shown in the studies mentioned. In studies that focus on the printing time for different filling patterns, such as [23], the built-up direction is not taken into account. This is where the present study comes in, examining a wide range of infill densities for different patterns in all three built-up directions and including printing time considerations.

The examined patterns in this work are limited to cubic and triangular filling patterns, which are shown schematically in Figure 2. Increasing the infill density generally improves the mechanical performance, although this comes at the cost of longer print times and higher material consumption [24]. Hence, the optimization of the internal architecture is a balancing act between mechanical strength and manufacturing efficiency. In terms of the used material, poly lactic acid (PLA) remains one of the most commonly used thermoplastic materials in FFF due to its biodegradability, ease of processing, and favorable surface finish [2,25]. However, PLA exhibits relatively brittle failure behavior and sensitivity to print parameters, especially in vertical built-up directions where interlayer adhesion is

weakest [4,6,9]. Despite the extensive literature on individual process parameters, systematic investigations addressing the combined influence of infill density, infill pattern, and built-up direction on the flexural strength of FFF-manufactured PLA components remain limited. Understanding these relationships is of high practical relevance, as FFF technology continues to expand into functional and load-bearing applications in mechanical engineering, biomedical devices, and lightweight structural components [26,27].

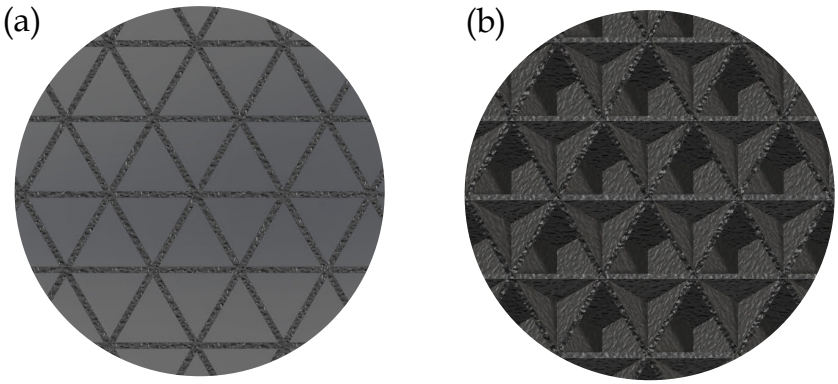


Figure 2. Scheme of the investigated infill patterns: (**a**) triangular; (**b**) cubic.

For example, components printed using PLA are being investigated in [28] for toolmaking, such as bending tools for processing thin metal sheets. Furthermore, there have already been trials of prototype cutting tools made from PLA, which were tested on aluminum sheets [29]. In the field of medical technology, PLA components are also increasingly being used in the form of bone scaffolds, which are combined with bioactive fillers [30].

Identifying the optimal interplay between material utilization, print time, and mechanical performance can guide process parameter selection for enhanced reliability and sustainability. Therefore, the present study aims to experimentally investigate the influence of infill density, print pattern, and all three Cartesian built-up directions on printing time and flexural strength of FFF-printed PLA samples through standardized three-point bending tests according to ISO 178 [31]. By integrating all these parameters within a unified experimental framework, this study advances the current state of knowledge on the flexural performance of 3D-printed PLA samples.

The results are intended to provide quantitative insights into parameter optimization for improved design strategies and predictive modeling of anisotropic behavior in additively manufactured polymers.

2. Materials and Methods

2.1. Experimental Setup

The experimental method of finding the flexural strength of PLA samples in this paper is based on ISO 178 [31]. This document specifies a method for determining the flexural properties of rigid and semi-rigid plastics under defined conditions. The method is used to investigate the flexural behavior of the test samples and to determine the flexural strength, flexural modulus, and other aspects of the flexural stress/strain relationship under the conditions defined. It applies to a freely supported beam, loaded at midspan (three-point loading test), according to Figure 3. In principle, a test sample of length l and a rectangular cross-section (width b, height h) resting on two supports is deflected by means of loading edge action on the sample midway between the supports.

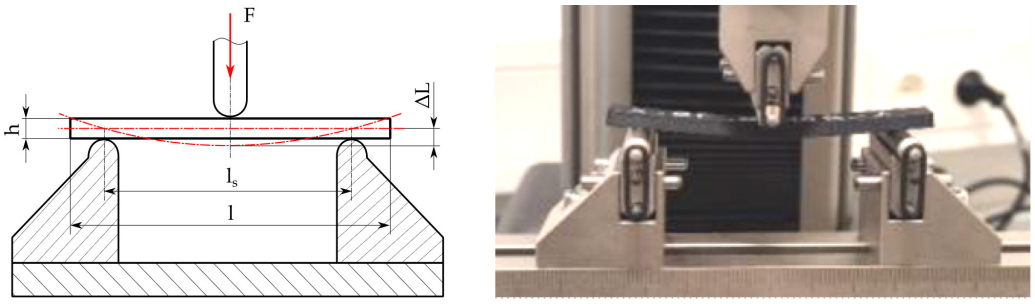


Figure 3. Left: Scheme of bending test principle. **Right**: Real application of the bending test.

The preferred sample calls for a 2 mm/min test speed, which was used in this case, according to [31]. The test sample is deflected in this way at a constant rate at midspan until rupture occurs at the outer surface of the sample or until a maximum strain of 5% is reached; whichever occurs first. During this procedure, the force F applied to the sample and the resulting deflection ΔL of the sample at midspan is measured. Based on the gained data, material properties as flexural stress

$$\sigma_f = \frac{3Fl_s}{2bh^2}, \tag{1}$$

and flexural strain

$$\epsilon_f = \frac{6\Delta Lh}{l_s^2}, \tag{2}$$

were finally calculated. The Young modulus could also be evaluated with

$$E_f = \frac{\sigma_{f2} - \sigma_{f1}}{\epsilon_{f2} - \epsilon_{f1}}, \tag{3}$$

where σ_{f1} and σ_{f2} have been chosen at $\epsilon_{f1} = 0.0005$ and $\epsilon_{f2} = 0.0025$, respectively, according to [31]. The flexural tests have been terminated at 5% strain [31], which, using Equation (2), translates to a displacement of approximately 8.53 mm for a sample with a thickness of 4 mm and a support span of 64 mm. For this study, all samples were tested under ambient laboratory conditions. Samples, which were printed in X- and Y-directions (see Section 2.2 for labeling scheme) were intentionally bent beyond the 5% strain limit in order to obtain a more comprehensive dataset, which could be utilized for further analyses concerning the failure methods of PLA prints, which are examined in Section 3.3.

This overbending has no influence on the measured maximum values of the bending stresses, nor on the comparability of the different built-up directions. A displacement limit of 4 mm was set for samples printed in the Z-direction due to the increased likelihood of premature failure caused by weak interlayer adhesion [9,24,32] as well as observations from preliminary tests, which consistently showed brittle fracture behavior before this threshold. Data acquisition was carried out continuously throughout the test, capturing force-displacement values which were later used to find the maximal flexural strength.

2.2. Material, Sample Geometry, and Printing Orientation

The material used for all prints was PLA from manufacturer Polymaker with a filament diameter of 1.75 mm. The geometry of the used samples chosen was cuboid, according to the standard [31], with dimensions and allowed deviations as listed in Table 1. More specific data regarding material and slicer settings can be found in Table 2.

Table 1. Suggested sample dimensions, according to ISO 178 (dimensions in mm).

Length l	Width b	Height h
80 ± 2	10 ± 0.2	4 ± 0.2

In any of the test samples, the thickness within the central one third of the length shall not deviate by more than 2% from its mean value and the width shall not deviate by more than 3%. The samples have been measured using a micrometer gauge, within the measurement ranges, to the nearest 0.1 mm for the width and to the nearest 0.01 mm for the thickness. The mean values of thickness and width are to be calculated for each set of test samples and any samples exceeding the aforementioned tolerances were discarded and replaced by another sample. It was also assured that the samples were all free of twist, sink marks, scratches, pits, and flash and preferably have mutually perpendicular surfaces. In order to be able to assess the influence of the layer orientation as well as the infill structure on the flexural strength, samples were produced in three different build-up directions.

These are depicted in Figure 4, in accordance with the defined coordinate system. The colored lines on the depicted sample additionally define the layering relative to the applied bending load. Furthermore, for every build-up direction, samples with different infill density (10%, 30%, 50%, 70%, and 90%), along with two different infill patterns, namely triangular and cubic, were produced. Since at least five samples shall be tested in each configuration, the total amount of samples manufactured and tested was 150 (one hundred fifty). Figure 4 also shows the labeling scheme that was defined to clearly differentiate all types of samples produced. A vertically (Z-direction) printed sample with a cubic infill pattern and an infill density of 30% is labeled with Z-C-30%, for example. For later comparison, additional samples with a 100% infill density printed in all directions were measured using the same method as described.

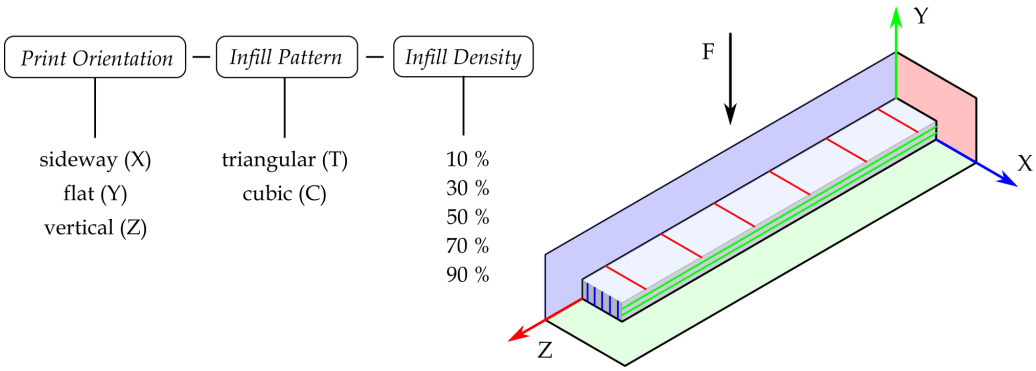


Figure 4. Left: Test series combinations and labeling scheme for test samples. **Right**: Definition of the coordinate system for built-up direction specification.

2.3. Printing and Slicing Settings

For the production of all samples in this study, a bed slinger equipped with a Bowden-style extruder 3D-printer from Prusa was utilized. To ensure consistency and reproducibility, all slicing parameters were standardized, and adjustments were only made where necessary for specific orientations or infill variations. Filament was extruded at the recommended temperature as per the manufacturer's guidelines. Samples were manufactured at a 0.2 mm layer height, which is standard for 0.4 mm diameter nozzles. Two perimeter layers were printed for the completely filled margins on the upper and lower sides of the samples. This number was selected to minimize the influence of solid layer behavior in the mechanical analysis while maintaining flat and uniform faces, as demanded by the standard. For prints along the Z-axis, a three-layer raft was employed to improve bed

adhesion and mitigate the risk of print failure due to reduced contact area. The used settings, as well as relevant material data, are summarized in Table 2.

Table 2. Summary of printing and slicing settings.

Printer	Prusa Mini (Firmware 6.4.0-RC2+11973)
Slicer	Prusa Slicer Ver. 2.9.4
Filament Manufacturer	Polymaker (Changshu, China)
Filament Material	PLA (PolyTerra)
Filament Diameter	1.75 mm
Filament Color	Charcoal Black
Bed Temperature	60 °C
Nozzle Temperature	215 °C (230 °C for first layer)
Nozzle Diameter	0.4 mm
Layer Heigth	0.2 mm
Extrusion Multiplier	1
Printing Speed (Infill)	115 mm/s
Printing Speed (Perimeter)	45 mm/s
Perimeters	2

3. Results

3.1. Flexural Strength

Figure 5 illustrates the effect of varying infill density on the flexural strength of samples printed with different infill patterns and orientations. The data reveal a general trend of increasing flexural strength with higher infill density across all configurations. The highest strength values were achieved by the cubic configurations, both exceeding 30 MPa at 90% infill, indicating superior mechanical performance of the cubic pattern in both horizontal orientations independently. In contrast, *Z*-oriented samples consistently exhibited the lowest flexural strength as expected, highlighting the negative impact of vertical layering on mechanical integrity. The weakness of the structure in the *Z*-direction is clearly visible. While the flexural strength for the X- and Y-built-up directions increase steadily regardless of the filling pattern, the bending strength for Z-printed samples decreases even at the highest filling level compared to 70% infill density.

Both X-printed configurations also displayed high performance at low infill densities, with a noticeable plateau in strength gains beyond 50% infill. It is assumed, that this plateau observed is a reason of change in the characteristics of the material's inner support structure.

For samples with triangle infill, a clear difference can still be seen, even at 90% infill density, which is not the case for the cubic pattern, where values of flexural strength merge. The two curves already start to converge between 50 and 70%, although it should be noted that this is due to a flattening of the cubic curve. In general, the cubic pattern leads to increased flexural strength compared to the triangular pattern within a build-up direction for X and Y. The opposite is true for the Z-direction, where the triangular pattern leads to higher values, regardless of the infill density. It is also interesting to note that the two curves for X- and Z-print directions run parallel over almost the entire area under consideration, whereas the two curves for Y-print direction diverge.

The flexural strength here increases noticeably faster for a cubic pattern with increasing infill density than with a triangular pattern. These prescribed findings underscore the critical role of both infill pattern and print orientation in optimizing the mechanical properties of 3D-printed components. In addition to Figure 5, the measurement results are summarized in Table 3, which also show results from measurements with 100% infill density.

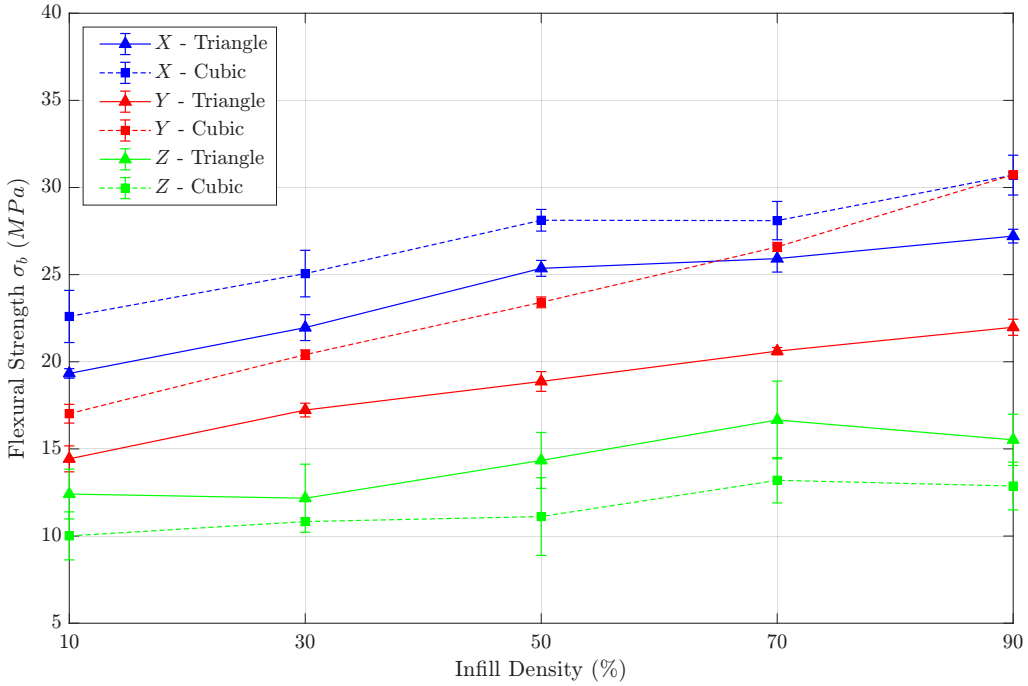


Figure 5. Flexural strength for different infill densities and patterns.

Table 3. Result data of flexural strength for different infill densities and patterns (mean value ± standard deviation).

	Flexural Strength (MPa)				
Infill Density	**10%**	**30%**	**50%**	**70%**	**90%**
X-Triangle	19.33 ± 0.27	21.96 ± 0.74	25.36 ± 0.45	25.92 ± 0.77	27.21 ± 0.39
X-Cubic	22.60 ± 1.49	25.06 ± 1.33	28.12 ± 0.62	28.10 ± 1.10	30.71 ± 1.14
Y-Triangle	14.43 ± 0.74	17.23 ± 0.39	18.87 ± 0.56	20.61 ± 0.21	21.98 ± 0.46
Y-Cubic	17.02 ± 0.53	20.41 ± 0.27	23.41 ± 0.31	26.59 ± 0.26	30.73 ± 0.13
Z-Triangle	12.41 ± 1.42	12.17 ± 1.95	14.34 ± 1.60	16.66 ± 2.23	15.52 ± 1.47
Z-Cubic	10.01 ± 1.37	10.83 ± 0.21	11.12 ± 2.23	13.20 ± 1.30	12.87 ± 1.37
Infill Density			100%		
X-Rectilinear			43.46 ± 2.98		
Y-Rectilinear			37.15 ± 0.83		
Z-Rectilinear			22.57 ± 2.47		

3.2. Printing Time

To better illustrate the relationship between mechanical performance and production efficiency, Figure 6 plots flexural strength against printing time for all X and Y configurations across varying infill densities. For better presentation, the results have been rounded to whole minutes.

Samples printed in the Z-direction were not considered here due to their low relevance (printing times of up to 47 min). The trend lines clearly show that cubic samples in the Y-print direction (Y-C) achieve the highest flexural strength with a simultaneously very low printing time of 11 min for 90% infill density, highlighting it as the most efficient configuration in terms of strength-to-time performance.

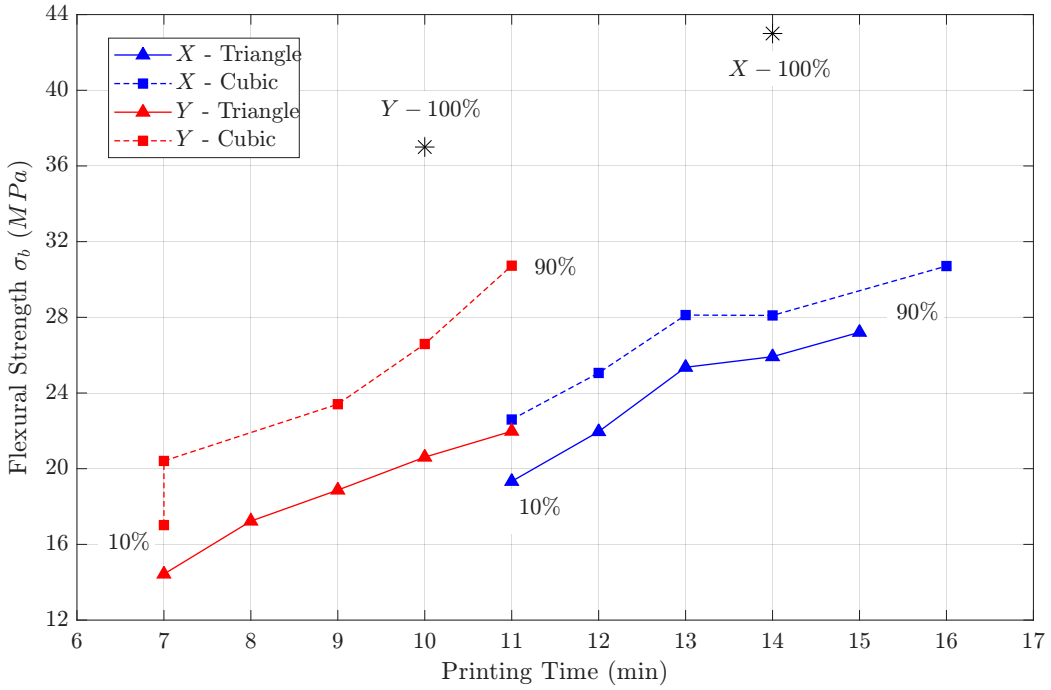


Figure 6. Flexural strength as a function of printing time.

While cubic samples printed in the X-direction (X-C) reach comparable strengths, they do so with an approximately 30% longer print time. This visual comparison supports the conclusion that Y-C configuration provides the optimal balance between print time and mechanical strength, making it a highly favorable option for time-sensitive yet performance-driven applications, such as prototyping, for example.

Although results from 100% infill density are not directly comparable due to different patterns, they show an increase in flexural strength of up to 41% compared to both C-90% configurations at lower printing times.

3.3. Failure Methods

During mechanical testing, several distinct failure mechanisms were identified, each correlating strongly with specific print orientations and infill configurations. In addition to the presentation of the possible types of failure that occur, the material behavior as a function of the printing direction becomes clear here, as already shown in [9]. The data from the respective configurations for which the type of failure was most obvious were selected for the illustration. The results, which are explained below, are summarized in Figures 7 and 8. The plotted curves in Figure 7 represent the mean values from five measurements carried out in each category, and Figure 8 shows the corresponding deformed samples.

3.3.1. Deformation Exceeded

Samples printed in X- and Y-directions mainly exhibited ductile behavior and sustained deformation beyond the 5% strain limit defined in ISO 178. These samples showed more gradual failure and continued to carry load beyond the initial yield point, indicating better energy absorption characteristics compared to Z-printed parts. In particular, samples printed in the Y-direction show high applicable loads against others, as shown in Figure 7.

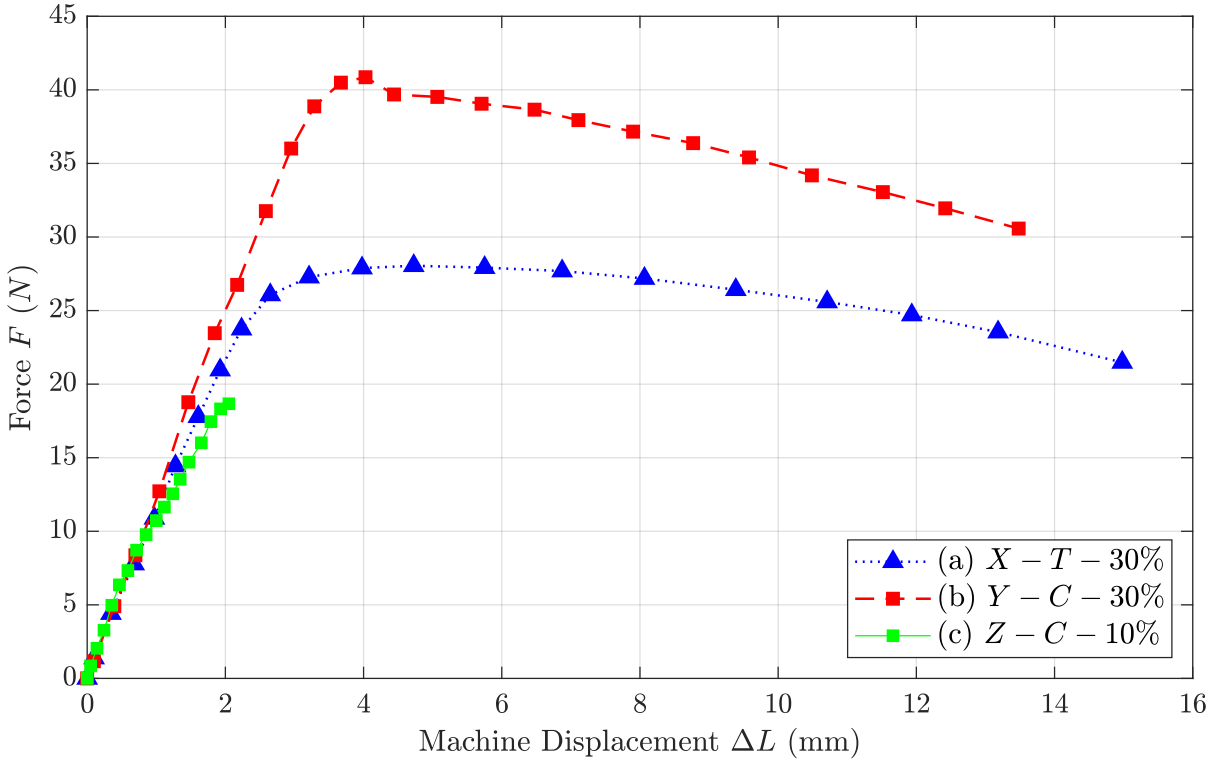


Figure 7. Data of testing failures characteristics.

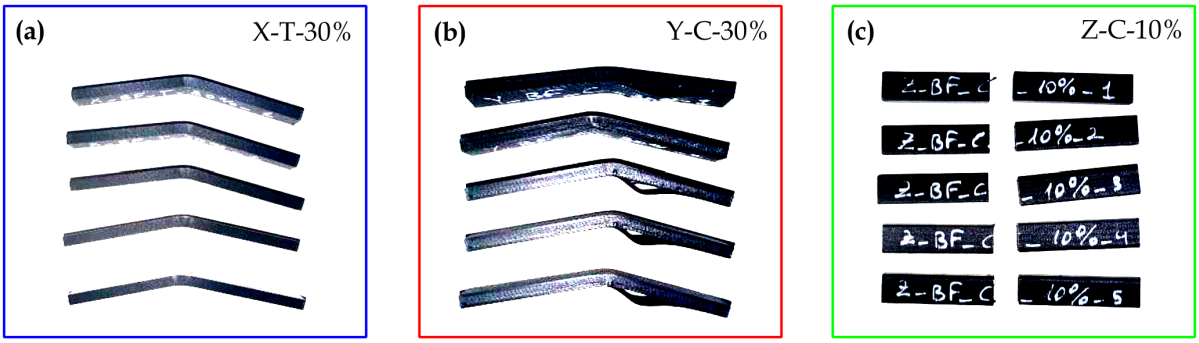


Figure 8. Sample deformation patterns for different printing orientations: (**a**) Ductile Behaviour, (**b**) Outer Wall Delamination, (**c**) Failure of Layer Adhesion.

3.3.2. Delamination of Outer Wall

In some cases, particularly at lower infill densities, samples showed signs of outer wall delamination, as can be seen in Figure 8b. This failure mode is often initiated at stress concentration points along the edges and propagated along the print lines. This is also clearly depicted in Figure 9 by at least one sharp, sudden drop in the applied force, followed by a gradual increase (areas in dashed circles), as the remaining layers and infill structure continue to support the load.

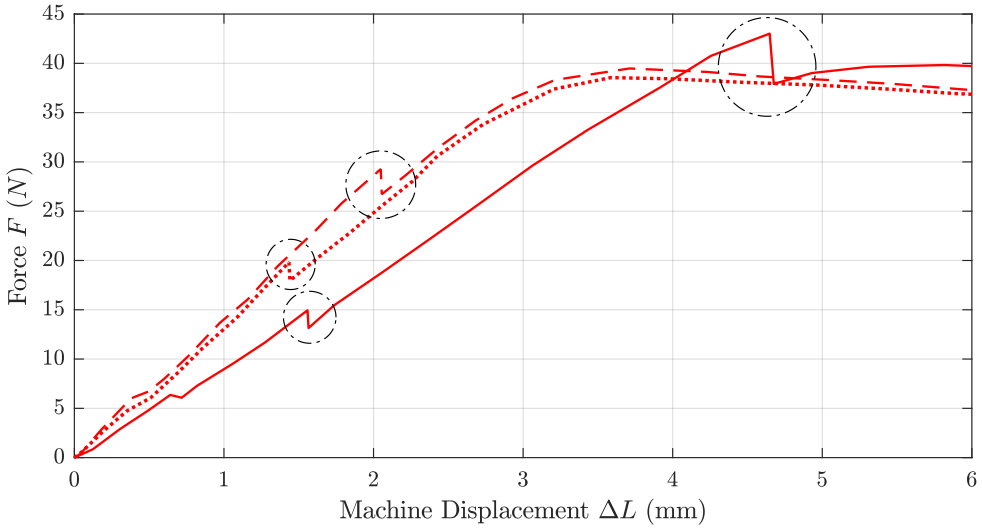


Figure 9. Force as function of printing time for three different samples in the Y-C-30% configuration (occurrence of wall delaminations are marked with dashed circles).

3.3.3. Failure of Layer Adhesion

This was predominantly observed in all samples printed in the Z-direction. Due to the anisotropic nature of FFF printing, layer adhesion in the Z-axis is weaker compared to X- and Y-directions. As a result, the Z-oriented samples exhibited brittle behavior characterized by abrupt failure and steep stress–strain curves (Figure 7), despite relatively low force values. This confirms the susceptibility of vertically printed parts to delamination and sudden fracture, as the samples show in Figure 8c.

4. Discussion

The results of this study confirm that infill density, internal pattern, and built-up direction exert a significant influence on the flexural strength and failure behavior of FFF-manufactured PLA components. The findings align closely with previously published research and can be further substantiated by recent investigations focusing on mesostructural load transfer, local strain evolution, and fatigue behavior in polymeric and composite FFF parts.

4.1. Influence of Built-Up Direction

The pronounced anisotropy observed between horizontally (X/Y) and vertically (Z) printed samples corroborates the well-established directional dependency of FFF materials. Vertically built samples consistently exhibited lower flexural strength and brittle fracture behavior due to weaker interlayer adhesion and stress concentration along layer interfaces. Similar results have been reported by Song et al. [4] and Somireddy and Czekanski [6], who attributed the degradation of mechanical performance in the Z-direction to reduced molecular diffusion between layers and to void formation at the filament boundaries.

In addition, Gonabadi et al. [16] demonstrated that the reduction in strength in the Z-direction is not merely a consequence of poor adhesion but also of altered load-path geometry within the mesostructured infill. Localized shear and bending concentrations arise where infill filaments intersect at oblique angles, promoting premature failure. This mechanistic interpretation complements the experimental observations in this study, in which Z-oriented samples failed abruptly at lower stress levels.

Furthermore, the work of Rajkumar [19] shows that flatwise (Y-direction) and edgewise (X-direction) orientation reveal to have similar maximum flexural stress values at high infill densities, which underscores the findings in the present work.

4.2. Infill Density and the Transition from Infill- to Skin-Dominated Behavior

The nearly linear increase in flexural strength with infill density up to about 50% and the subsequent plateau observed in this work, especially for X-printed samples, can be explained by the gradual transition from infill-dominated to shell-dominated load bearing. Digital Image Correlation (DIC) analyses from recent studies [16,33] revealed that beyond a certain density threshold, stresses concentrate along the outer perimeters, while the infill mainly serves as a stabilizing core. This redistribution limits further strength gains despite the higher material volume, which is consistent with the measured flattening of the curves for both cubic and triangular X- and Z-patterns in this study.

4.3. Effect of Infill Pattern and Orientation Dependence

The superior performance of the cubic infill pattern in X and Y orientations can be attributed to its more homogeneous stress distribution and improved filament connectivity, leading to smoother stress–strain responses and higher maximum strength. Bonada et al. [33] demonstrated through both simulation and testing that orthogonal lattice structures exhibit nearly isotropic elastic moduli, whereas triangular or hexagonal infills create local stiffness peaks and strain gradients. These findings are in good agreement with the present results, where cubic infill yielded the highest flexural strength in horizontal orientations.

Conversely, for Z-printed samples, the triangular pattern occasionally outperformed the cubic structure. This inversion is likely due to the compact filament arrangement of the triangular lattice, which provides better inter-lamellar support and reduces the stress on weak layer interfaces, an effect previously noted for vertically printed composites by Ferreira et al. [25].

Regarding the examined infill patterns, a similar behavior consistent with the results of Rajkumar [19] is observed, where cubic samples outperform over triangular patterns in addition to others and tend to exhibit the highest maximum flexural strength.

4.4. Strength-to-Print-Time Efficiency and Fatigue Considerations

The Y–C configuration identified in this work as the most efficient compromise between mechanical strength and printing time confirms earlier optimization findings by Go et al. [24]. From an industrial perspective, such configurations are valuable for balancing productivity and performance in serial or functional prototyping.

The work of Padzic et al. [34] shows that when the fill density decreases from 100% to 90%, the tensile strength decreases by about 40%. Although both the type of load and infill pattern differ from the present work, a decrease in strength can be observed in approximately the same range.

The current work provides new insights, particularly in the area of the dependence of flexural strength on the build direction in combination with the infill pattern, under a 100% infill density. This is particularly relevant in the field of rapid prototyping. Functional components usually require higher permissible strength values, which necessitates the use of an alternative manufacturing process or an adjustment in the material, e.g., through carbon-fiber reinforcement.

However, recent research on carbon-fiber-reinforced PLA has highlighted that infill geometry and density also strongly affect fatigue life and crack propagation. Zhang et al. [35] demonstrated that gyroid or tri-hexagonal infills can increase fatigue endurance by up to 60% compared to simple cubic or rectilinear structures at similar densities. These results

suggest that future process optimization should extend beyond static flexural testing to include cyclic and dynamic loading conditions, especially for structural components.

4.5. Generalization Across Materials and Structural Reinforcement

The observed trends are not limited to PLA. Recent analyses on ABS and PETG parts [36] confirmed that anisotropy, infill-density thresholds, and orientation-dependent pattern effects are largely material-independent phenomena. This underlines the transferability of the findings to a wider range of thermoplastic feedstocks.

Moreover, Alagheband et al. [37] proposed adaptive infill strategies with locally increased density or reinforced shell–infill transition zones, which significantly delayed delamination and improved bending resistance in Z-built samples. Such variable-density or hybrid infill approaches represent a promising pathway to overcome the structural weaknesses identified in vertically printed samples of the present study.

5. Conclusions

The aim of this work was the analysis of influence from different combinations of built-up direction, infill density, as well as print patterns on the bending resistance of FFF-printed PLA-samples. Therefore, samples with 10, 30, 50, 70, and 90% infill densities were printed with cubic and triangular patterns in all three possible built-up directions and examined under a standardized three-point bending test. The findings from the study can be summarized as follows:

- Printing in the Y-direction with a cubic pattern and infill densities above 60% provides an optimal compromise between flexural strength and production efficiency. With an infill level of 90%, the highest flexural strength value of just under 31 MPa was achieved for comparatively short printing times.
- Both X-printed patterns also displayed high performance at low infill densities. Up to an infill density of approx. 60%, the strength values for both infill patterns are even higher than those for the Y-C configuration. However, comparatively long printing times are a negative factor here.
- In general, it was possible to demonstrate the well-known fact that strength values increase with increasing infill density. Interestingly, this does not entirely apply to samples printed in the Z-direction in the present study. Here, the flexural strength at an infill density of 90% is at the same level (cubic) or even below (triangular) 70%.
- For samples printed in X- and Y-directions, the cubic pattern leads to increased flexural strength compared to the triangular pattern within the built-up direction, whereas for the Z-direction, the triangular pattern yielded higher values across all infill densities considered.
- Flexural strength for X- and Z-directions increase equally for both filling patterns as the infill density increases, whereas for samples in the Y-direction, flexural strength for the cubic pattern increases more strongly than for the triangular pattern.

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Abbreviations

The following abbreviations are used in this manuscript:

AM	Additive Manufacturing
ABS	Acrylnitril-Butadien-Styrol
DIC	Digital Image Correlation
FFF	Fused Filament Fabrication
MEX	Material extrusion
PETG	Poly Ethylene Terephthalate Glycol
PLA	Poly Lactic Acid

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